SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering ASSESSMENT TOOL 2- Short Answers

ITA0312
Course Code :

CO Assessed:
CO1

Weightage:
30% of CIA

CO1 Statement:

Student Name:

Section / Batch :

Code of Conduct
Course Title:
Mobile computing

Assessment:

Total Marks:
Assessment Tool 2- Short Answers

100 Marks

Analyze the *mobile* network components, mobile *base* station, mobile infrastructures *and*

protocols *in* cellular systems.

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Reg. No.:

Date:

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06/10/2026

(Name/Reg No) *certify that this* submission is *my original work and that I have*

adhered to *the guidelines* specified *for this assessment. I* understand that *any violation* of *academic integrity rules will*

result in *disciplinary* action.

Signature of the Student

Stakeens

Instructions

This test contains 50 questions, Total: 100 Marks.

Duration: 1 hour.

Faculty In-charge
Course Coordinator
Head of Department

Signature:

Date:
Signature:
Signature:

Date:
Date:

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning

team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10 log10 of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on

a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

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CO1-AT2- Short Answers

Total

Spectrum

45MHZ= 45,000KHZ

simplex channel = 25KHZ

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N =

45,000 2x25

45,000

900 full duplex

=

13

50

2)

Regional carrier

frequency Shared

N=

= 560

Shared = 7 cell

560

7

1)

80

3) A GSM operater carrier spacing = 200 KHZ = 100 carrier for each carring carring = 8

Time slots

N = 100 x 8 =

800

Voice channels.

4)

Base
station radiates
80w

hand se handset
transmit = 4W

P = 10

1910 (50)

4

10 log 10 (20)

= 10 13.011

Test

capture a serving

signal =-60dBM

Co-channel interference = -92 BM

C/2 = -60-(-92)

An operator

=-60+92

H

32 dB

holds spectrum = 42 MHZ = 42,000 KHZ

Assigning per channel=

frequencies = 7

N =
300KHZ

42,000
= 20
300 *X7*

7) Transmitter Power = 64w

exponent = 4
14
Path-loss

P2
=
24

8

Planner

want

ro

11

64

4

n

16

double a cell's racing radius = 2

P2 = 2 x P1

P2

=

471

cluster size = 4 voice channel = 15

N = 15x 4 = Go N=

10) =100

vechile traveling = 100 km/h = 27.78 *m/h*

camped

= 900 MHZ

fa = vf/*C*

27.78 × 900 x 106 3x106

=

83.3 Hz